

SCALER

School of Technology

THE 15-CLASS FDE JOURNEY

Four phases. Fifteen classes.

PHASE 1 — DISCOVER

- C1 Stakeholder Discovery
- C2 Workflow + Problem Structuring
- C3 KPI + Scope

YOU ARE HERE — C2

PHASE 2 — DATA

- C4 Client Data
- C5 SQL + APIs + Files
- C6 Data Quality
- C7 Business Data Model
- C8 Dependable Pipeline

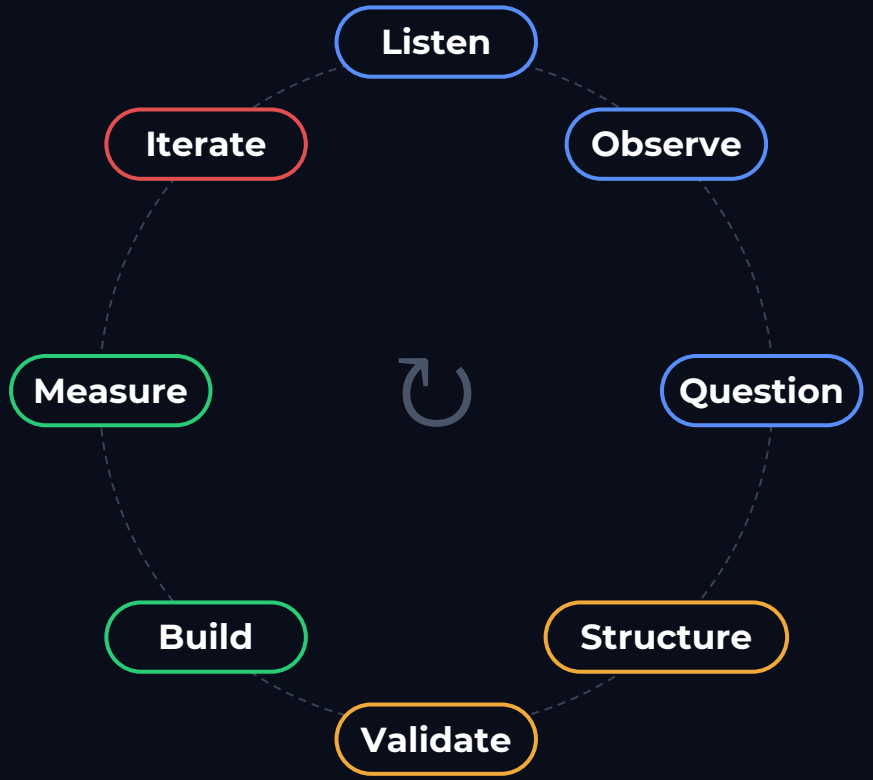
PHASE 3 — BUILD + AI

- C9 Product Wireframe
- C10 Agentic Coding
- C11 End-to-End Build
- C12 AI Agents
- C13 Multi-Agent + Evaluation

PHASE 4 — SHIP

- C14 Deploy + Observe
- C15 Client Review + Defence

THE FDE LOOP



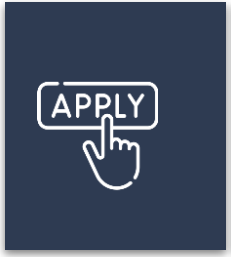
DON'T BUILD THE REQUEST.

DISCOVER THE PROBLEM.

Course objective



Understand the art of structured thinking
Learn how structured thinking enables defining and breaking down any problem to its nth level



Apply core problem-solving frameworks to real-world scenarios
Use proven methods to break down and tackle practical challenges with clarity and precision



Communicate your ideas clearly and logically
Present your thoughts in a way that makes your solutions easy to understand and follow

Structured Thinking: The Core of Problem Solving



*“**Structured thinking** is the **engine that drives problem solving**—without it, complex challenges remain chaotic and unsolved”*

How Structured Thinking Powers Effective Problem Solving



Breaking Down Complexity

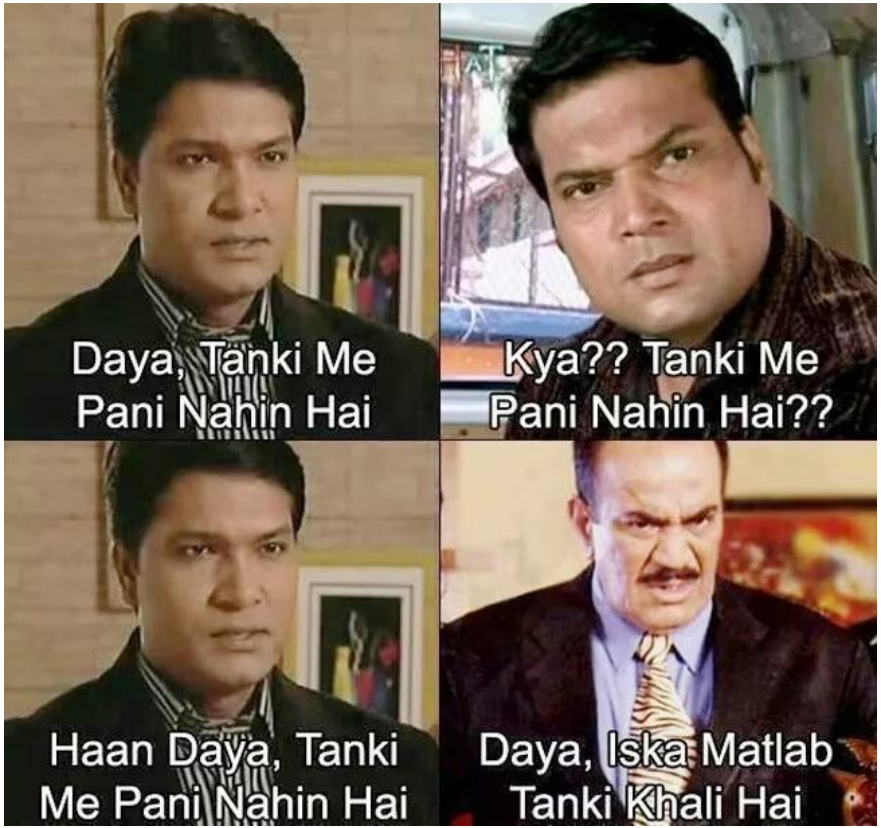


Creating Logical Flow

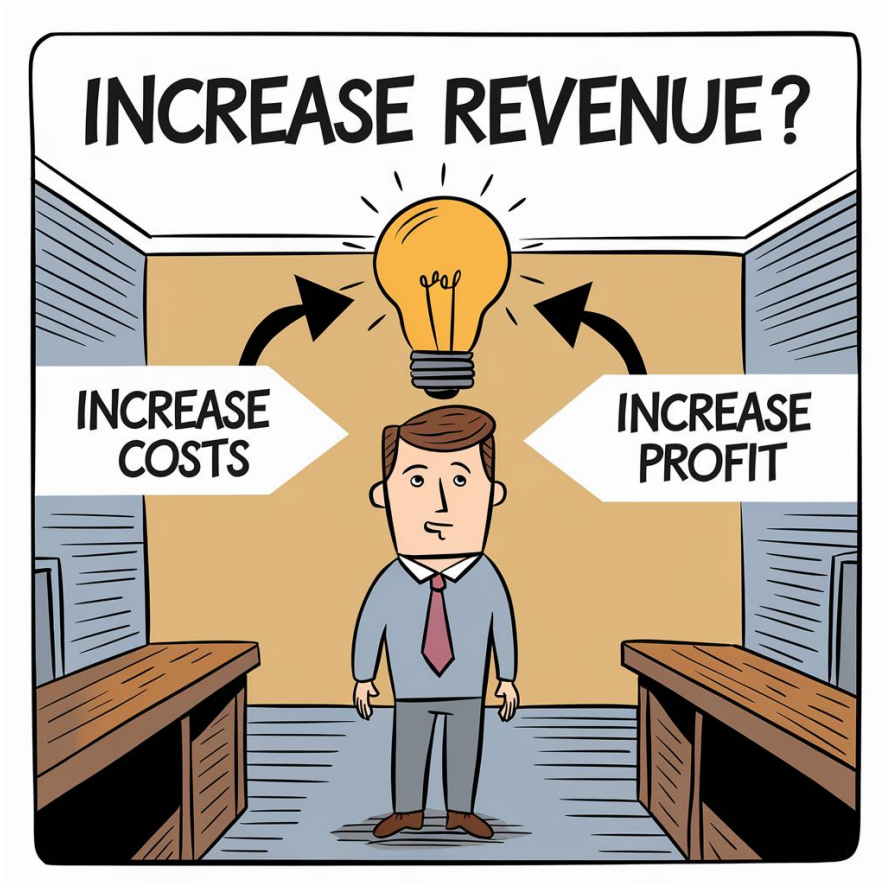


Focusing on Root Causes

What problem solving does not mean (1/2)?

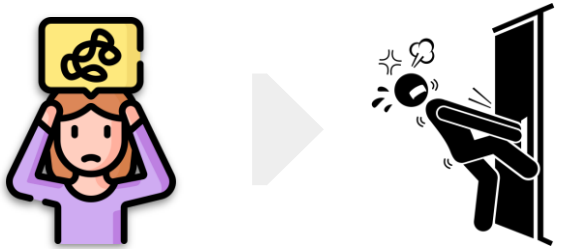


What problem solving does not mean (2/2)?



Problem solving - Example

Scenario: You have 5 minutes to unlock a door with 6 unique sets of locks and keys. What would you do?



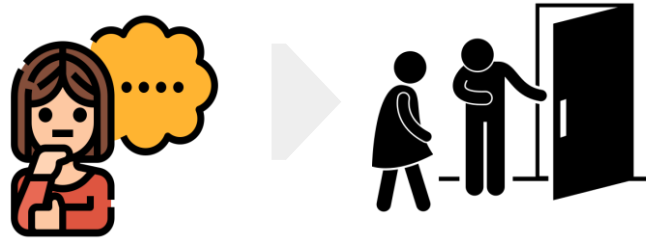
Anyone else



Tries each key randomly on every lock

Gets overwhelmed as time runs out

Repeats mistakes, hoping something will work



Problem solver



Observe the locks carefully

Sort the keys first

Test systematically

"Problem solving is the art of identifying root causes, analyzing options, and applying creative, strategic solutions. It requires clarity, adaptability, and a focus on long-term outcomes while addressing immediate challenges."

Pitfalls and challenges of general problem solving methodologies (1/2)

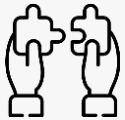
Major Pitfalls



Weak Problem Statements



Jumping to Conclusions



Not Breaking Down the Problem



Ignoring Team Input



Incomplete Analytic Tool Set

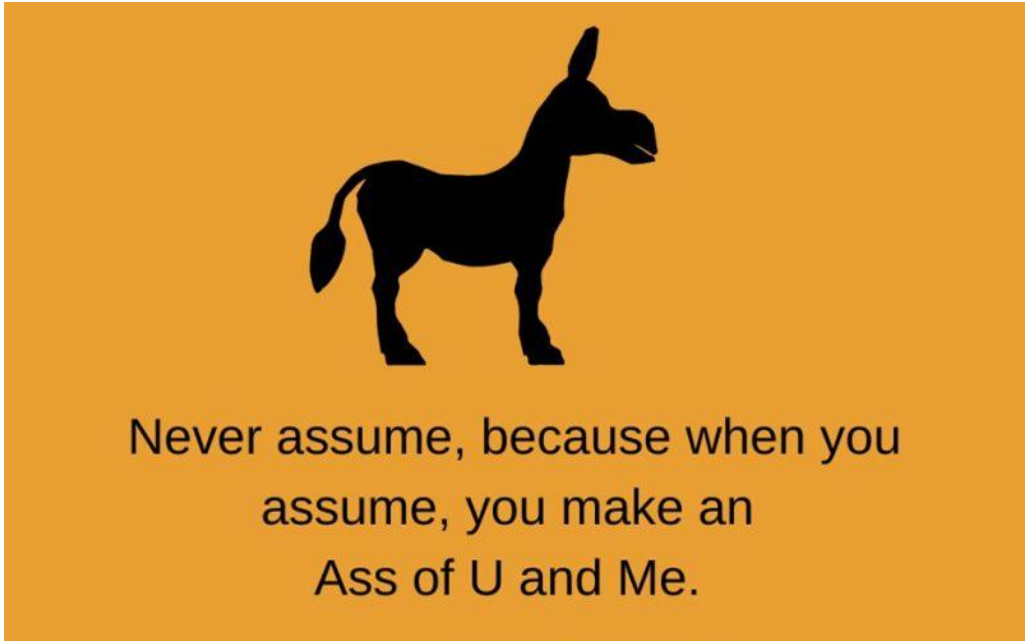


Treating Problem Solving as One-Off

...necessitate improvement in critical areas such as:

- Need for better problem definition
- Need for better structuring, prioritization & planning
- Need for better structuring & prioritization of issues
- Need for better planning & communication
- Need for better planning before conducting analysis

Pitfalls and challenges of general problem solving methodologies (2/2)



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Problem Solving Process

Cracking the Code: A Step-by-Step Guide to Effective Problem Solving

Defining the problem

What's the real issue?
Are we solving the root cause, or just a symptom?

Structuring the problem

How can we break the problem into MECE elements for maximum insight?

Prioritize issues

What key questions, if answered, will drive the most impact?
Don't get lost in the noise.

Plan analyses

What data do we need, ***how*** will we collect it, ***who*** is ***responsible, and what's*** *the* ***timeline?***

Conduct analyses

What's the quickest and simplest way to ***test the hypothesis?***

Synthesize findings

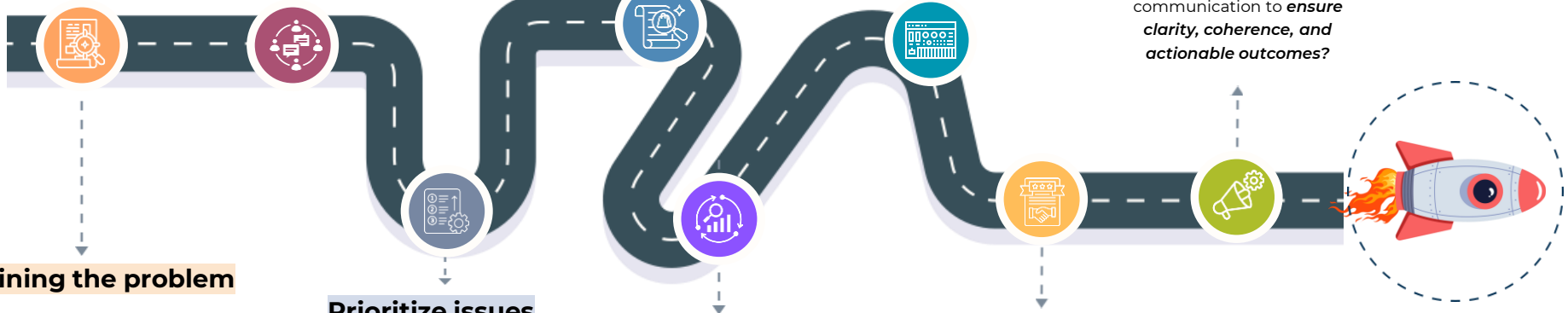
Define the "so whats?"

Craft Recommendations

What is the ***simplest, most effective solution, and why*** should we implement it?

Communicate

How can we structure communication to ***ensure clarity, coherence, and actionable outcomes?***

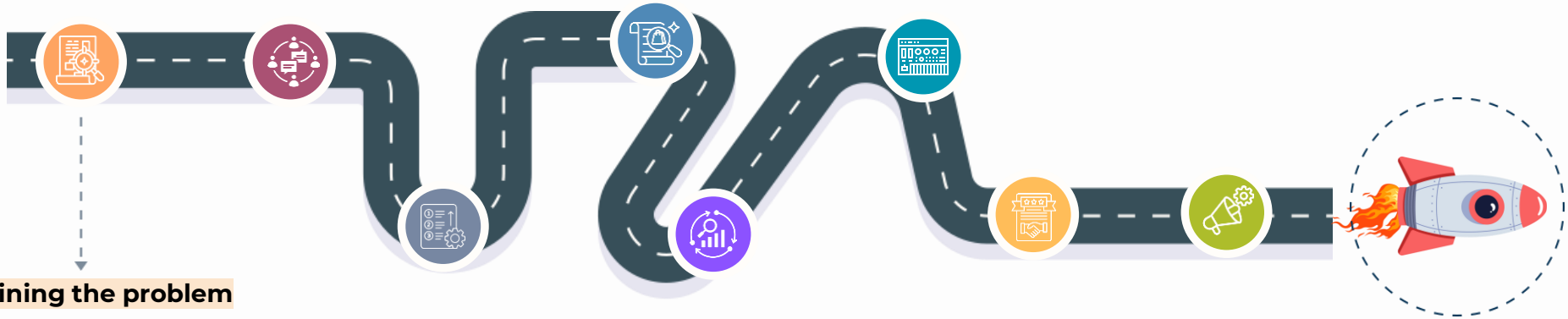


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Defining the problem

Initiating Our Problem-Solving Journey



Defining the problem

What's the real issue?
Are we solving the root cause, or just a symptom?

What does defining a problem statement mean?



At its core, defining a problem:

- **Pinpoints the core issue**
- **Creates Clarity**
- **Sets the Objective**
- **Narrows Focus**

"You can't solve what you don't fully understand. Defining the problem is about seeing beyond isolated parts—it's about capturing the entire picture. Without this clarity, every solution is just a guess."

Characteristics of a useful problem definition

The problem definition should be S.M.A.R.T - Specific, Measurable, Actionable, Realistic and Timebound

Specific	• Is the question specific enough to avoid ambiguity?
Measurable	• Can the outcome be measured to assess success / failure?
Actionable	• Does the question lead to tangible measures that can be actioned?
Realistic	• Is what you are solving genuinely achievable?
Timebound	• Is there a clear timeframe?

Scenario- Website checkout (1/2)

Scenario: Your client is an e-commerce retailer that has noticed a drop in completed purchases at checkout. Identify the reasons and recommend solutions

How to define the problem statement?

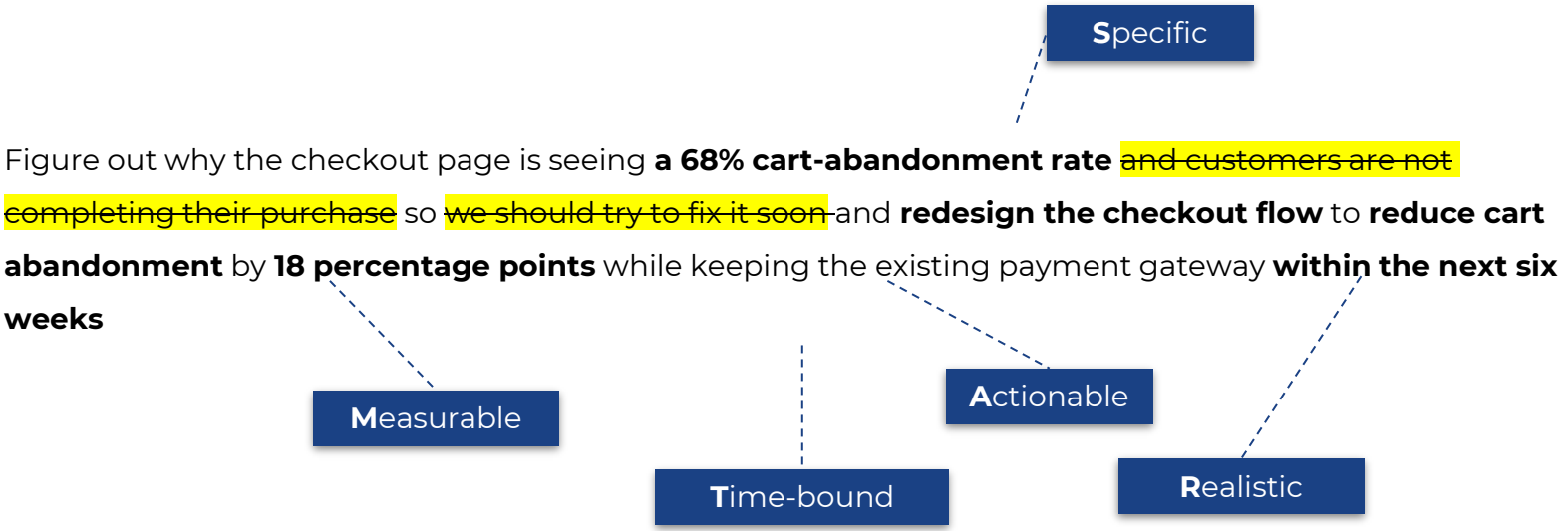
Figure out why customers are not completing their purchase and try to do something about it soon

Vague & Ambiguous

- What exactly is the **problem?**
- What **metric** company wants to improve?
- What is the **time-frame?**

Scenario- Website checkout (2/2)

How to define the problem statement?



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Structuring the problem

Pause and reflect

Structuring the problem

How can we break the problem into MECE elements for maximum insight?



Defining the problem

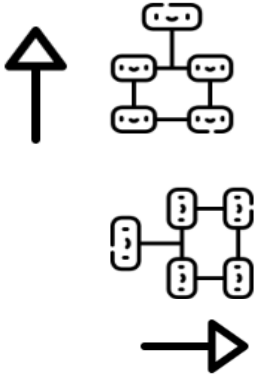
What's the real issue?
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Logic Trees: Unpacking Problems Step-by-Step

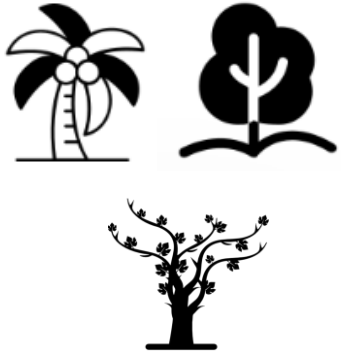
The Tree Analogy: A Powerful Tool for Organizing and Decoding Problems



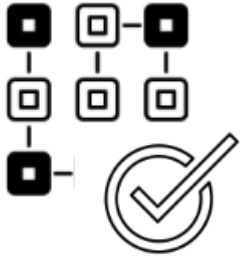
Structure for Organizing



Visual Tool



Multiple Approaches



Mental Model

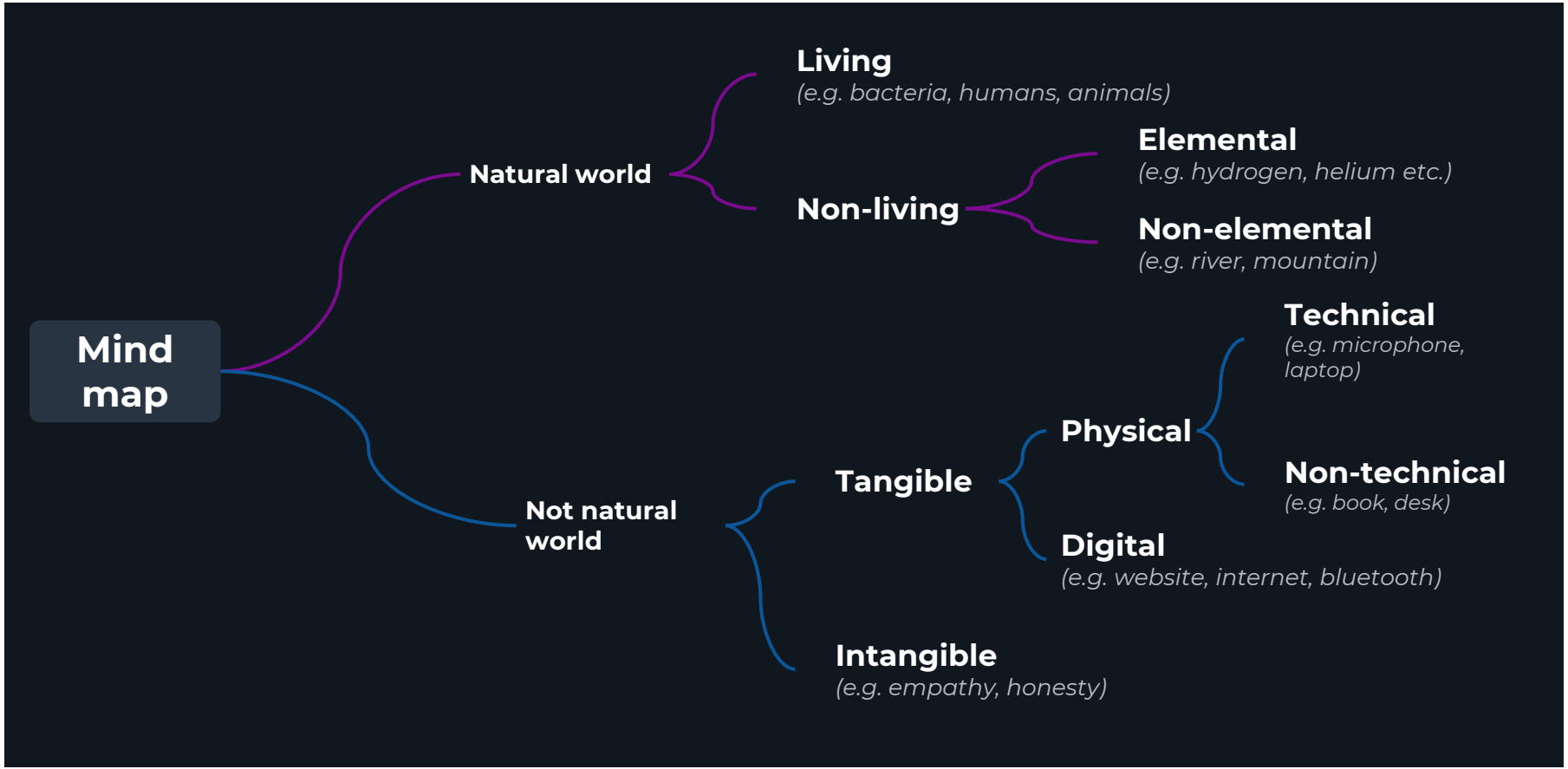
"You can't solve what you don't fully understand. Defining the problem is about seeing beyond isolated parts—it's about capturing the entire picture. Without this clarity, every solution is just a guess."

Why is it needed?

Let’s play a game



Example of logic tree



In summary, a logical tree,

- **Simplifies Complex Problems:** Breaks down large problems into smaller, clearer parts
- **Visual Structure:** Provides a clear, organized framework for decision-making
- **Improves Focus:** Helps prioritize and address key issues systematically
- **Encourages Comprehensive Solutions:** Ensures all elements of the problem are considered
- **Supports Actionable Steps:** Guides towards specific actions based on the breakdown

“KYT”: Know your trees

Navigating Rameshwaram Café’s Growth using Logic Trees

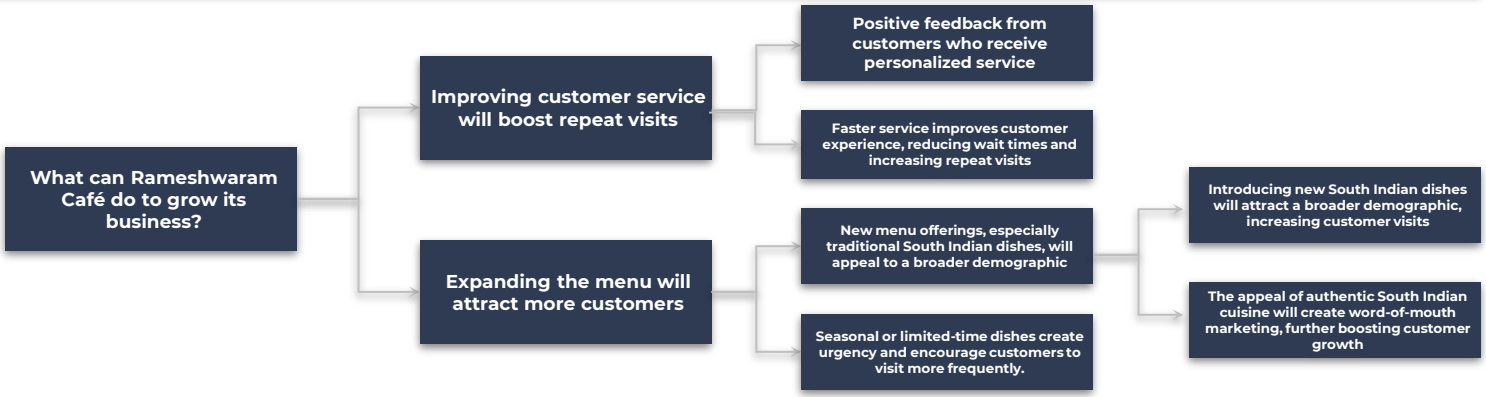
1 Issue tree

When to Use: Early on, when you don’t know much about the underlying structure or can’t yet form hypotheses



2 Hypothesis tree

Scenario: When you know enough about the structure to make clear hypotheses to begin to test with your data gathering



Characteristics of Issue & Hypothesis trees: Mapping Problems vs. Testing Solutions

Issue Tree

What could be causing this problem?

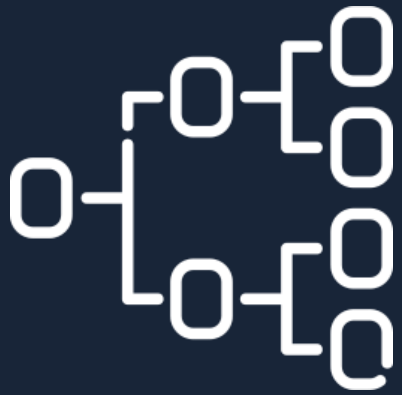
- Sub-branches break down each issue into sub-elements
- should be mutually exclusive and collectively exhaustive (MECE)

Hypothesis tree

Is this explanation correct, and how can we test it

- Is it testable - Can you prove /disprove it?
- Is it open to debate, or just a fact?
- If reversed, does it impact your logic?

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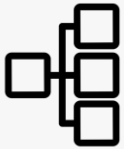


Issue Trees

The Issue Tree [Mapping Problems]

An issue tree transforms complexity into clarity, breaking down problems into manageable parts so that solutions become visible

What is an Issue Tree?



Maps out potential reasons



Should be MECE

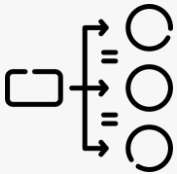
When use an Issue Tree?



Lack of knowledge to form any hypotheses

Key Principles to build an Actionable & Impactful Issue Tree (1/2)

Master the essential principles to construct a clear, actionable, and impactful issue tree that drives effective problem-solving



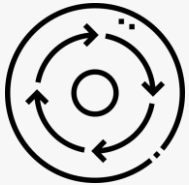
Parallel structure



Actionable



Focus on Root Causes



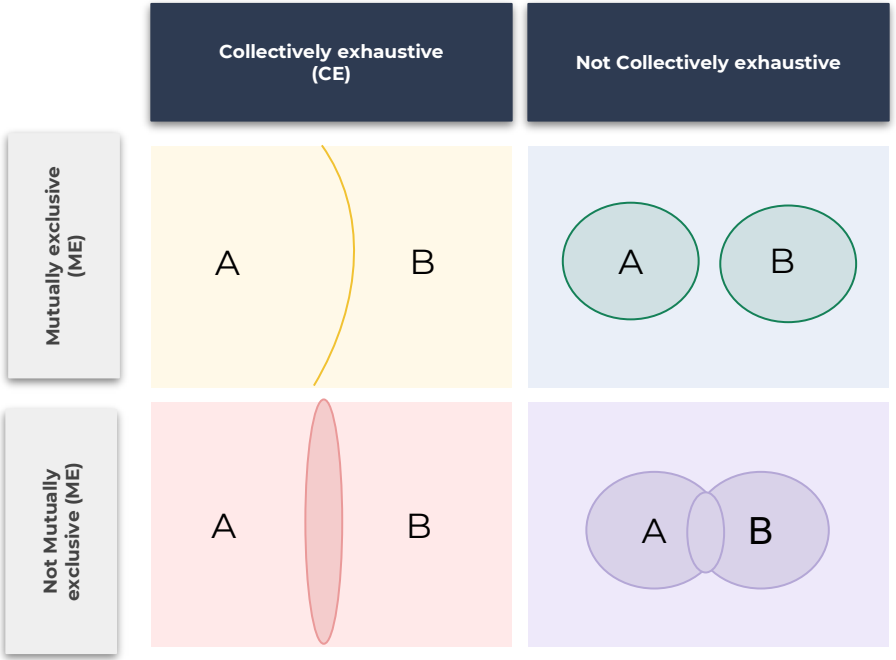
Consistency



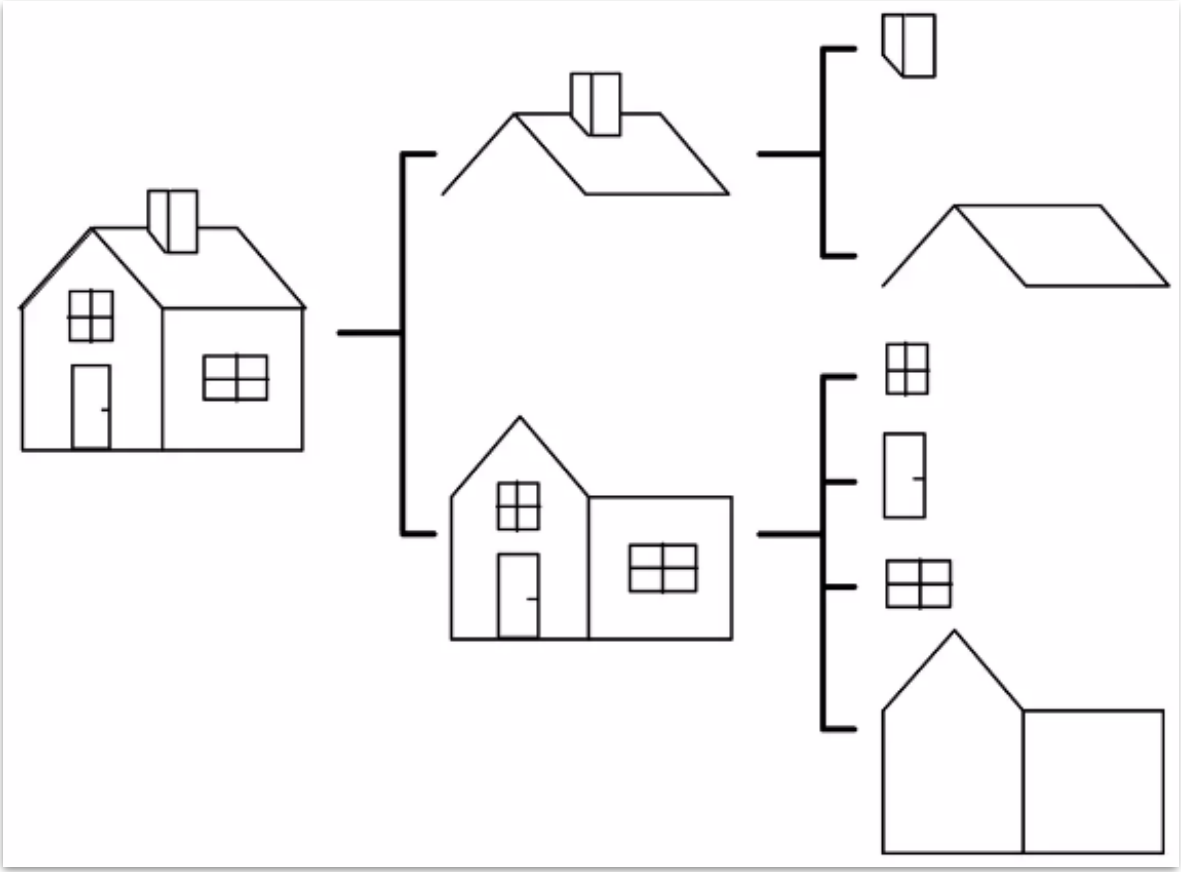
Should be MECE [Mutually Exclusive and Collectively Exhaustive]

Key Principles to build an Actionable & Impactful Issue Tree (2/2)

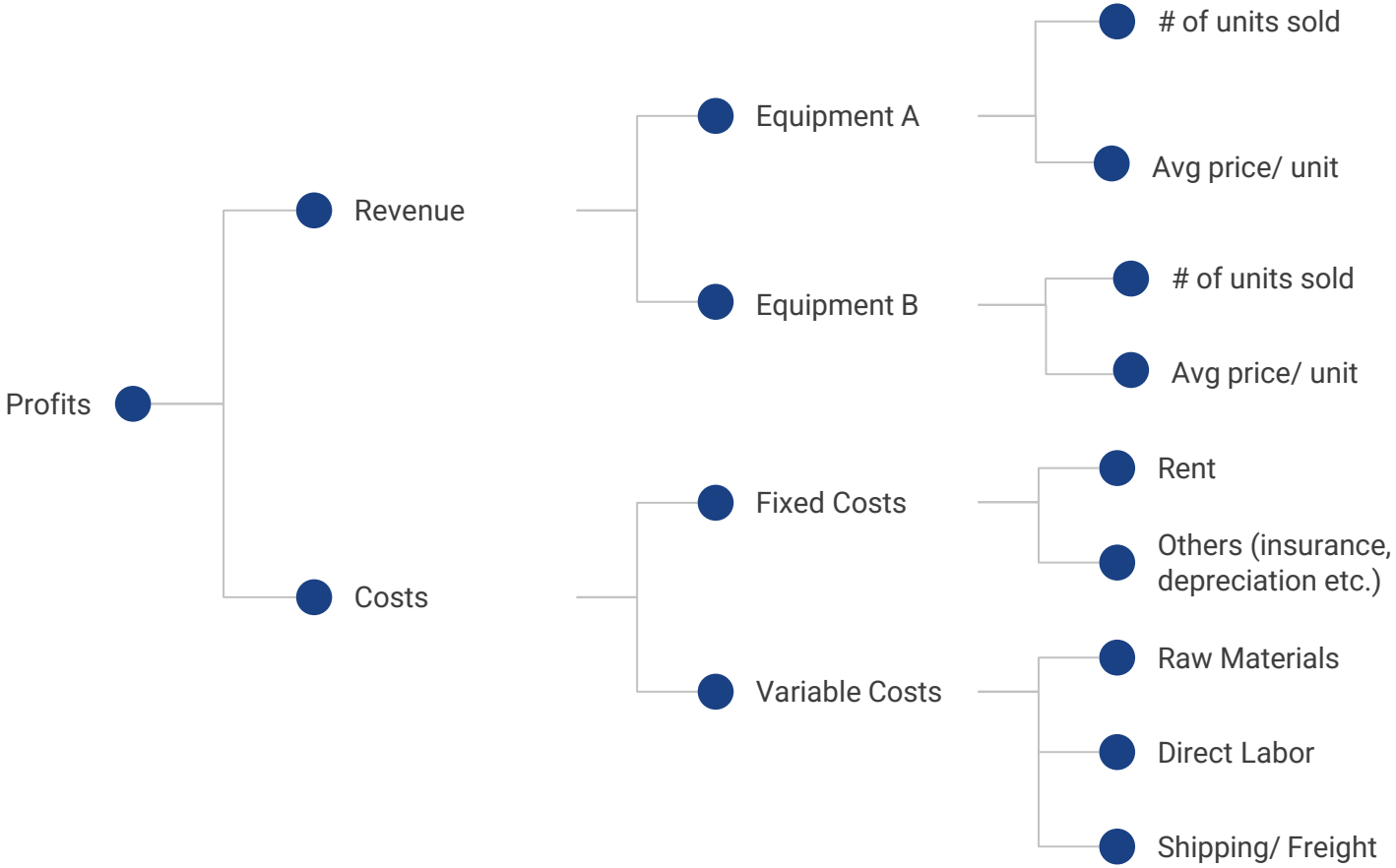
This principle helps in problem-solving by making sure you're not missing anything (CE) and not repeating yourself (ME): Mutually Exclusive and Collectively Exhaustive (MECE)



Structuring the Problem (1/2)



Structuring the Problem (2/2)



Let us solve a puzzle



You are a **technical engineer** with expertise in TV / Fridge Repair working in a home services company situated on the planet **Mars**, called **Rural Company.**

A customer has raised a request that **the light in their fridge is not working.**

How would you check whether or not the bulb is working?

<u>One difference in technology on Earth vs Mars:</u>
The bulb glows when the door is shut, rather than when its open

Structured Approach

Emits Light	Generates Heat	Consumes electricity
Install a camera inside	Touch it & check for warmth	Measure electricity consumption with and without the bulb
Put a recording device	Put heat sensitive material inside	Measure change in resistance using a Multimeter across the bulb
Put “photo” sensitive material inside like a photo film	Bulb’s filament should be elongated	

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Prioritize Issues

Pause and reflect

Structuring the problem

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Defining the problem

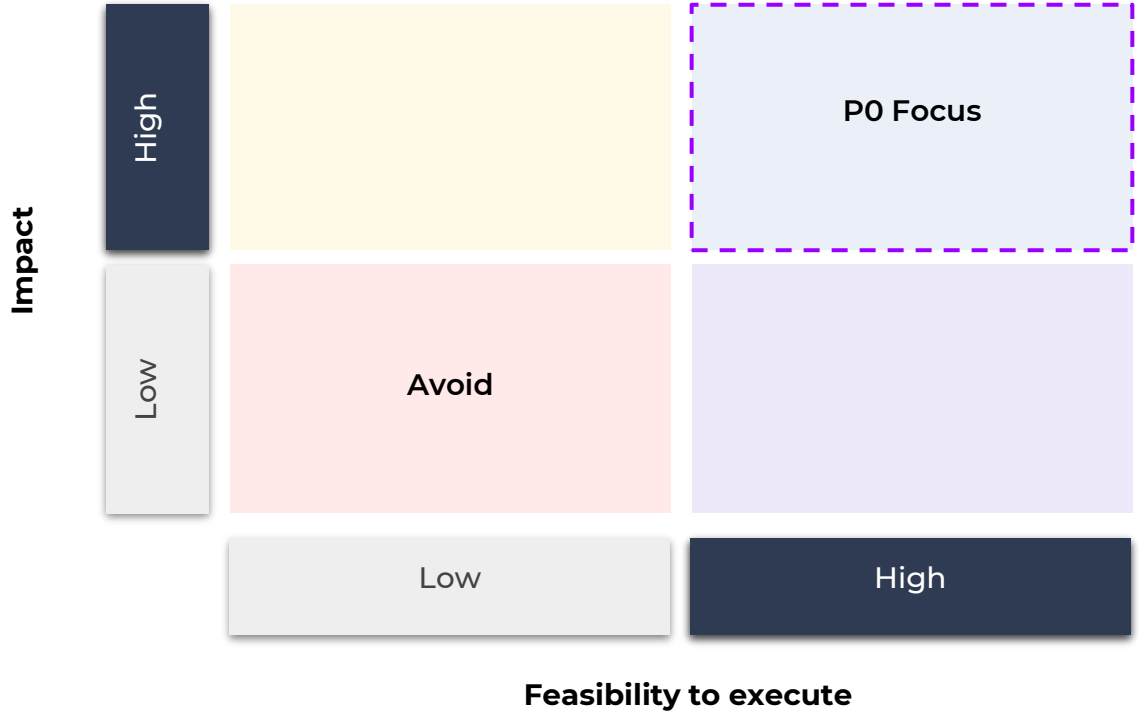
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Don't get lost in the noise.

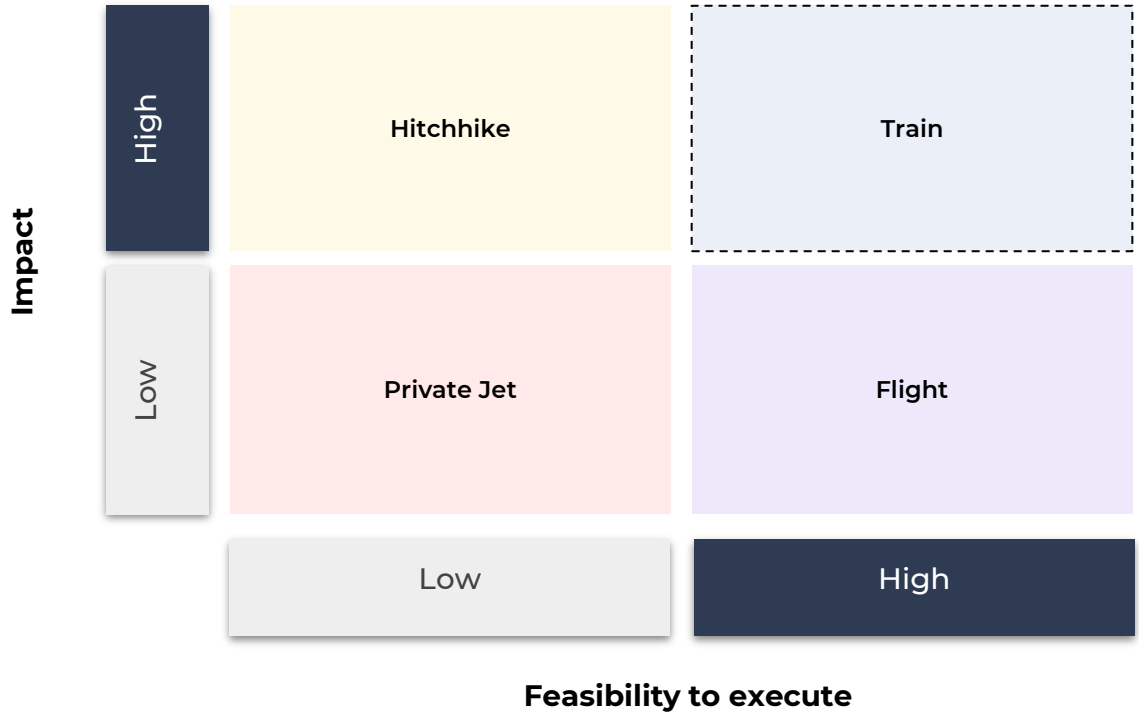
Prioritize and Prune: Focusing on What Matters Most

With limited resources, focus on the most feasible issues that offer the highest potential impact on the final outcomes



Prioritize and Prune: Cost effective travel

Problem statement: *How can we travel from Bangalore to Delhi in a cost effective manner (we can't walk there for sure)*



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Plan Analyses

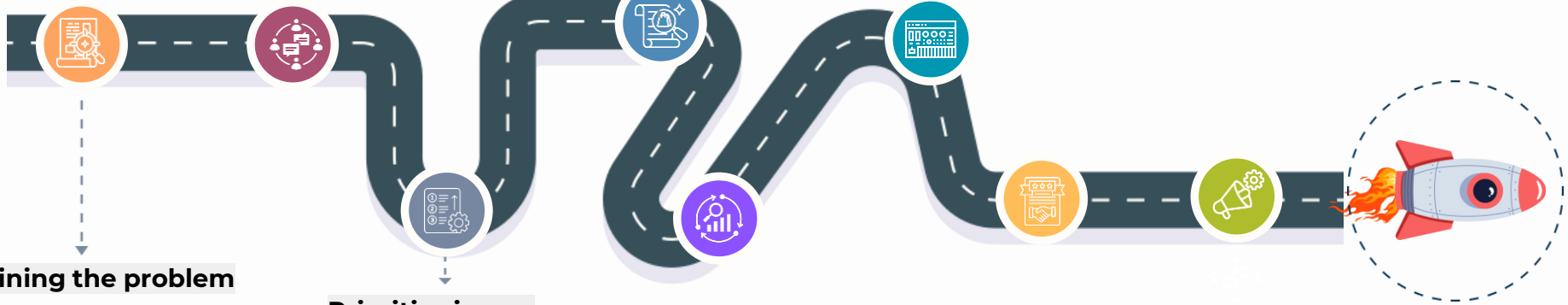
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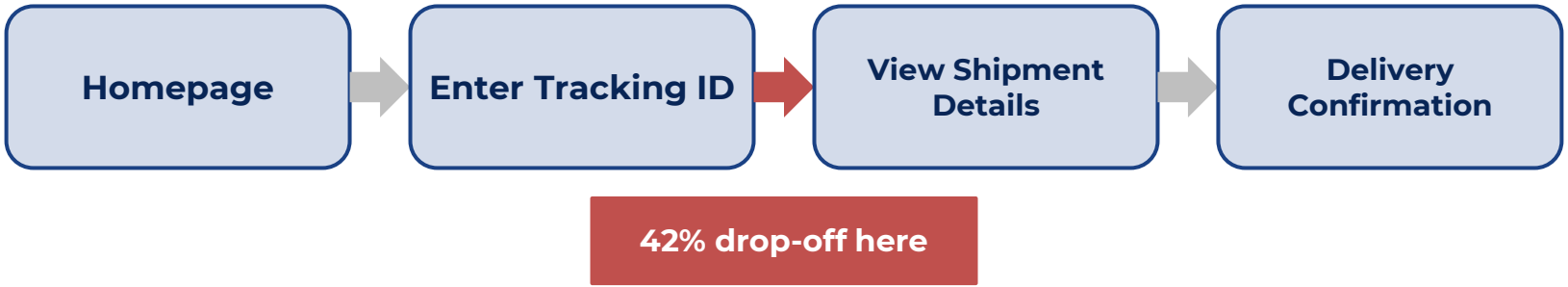
What key questions, if answered, will drive the most impact?
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Plan analyses: Template

Issue	Hypothesis	Analysis	Sources of insight	End product	Timing /steps / responsibility
• Starts with endpoints (or 'leaves') from logic tree. • The definition of an issue varies from an important question to an unresolved question. • It is phrased so that it can be answered yes or no and on which a specific action depends	The hypothesis is a statement of the likely resolution of the issue. It includes the reason for answering yes or no List all hypotheses: • Front-line ideas • Own ideas • Colleagues' ideas • Discuss with team members • Refine hypotheses • Re-adjust priorities for analysis	Indicate what analysis will be done to provide insights that would answer the question or test the hypothesis	• What sources will we use to generate the necessary insights? • If we need data, where will we get it? Is the data available internally? Can we buy the data? Get it free off the Internet? • If we need to interview experts, how will we find them? • If we need to conduct a survey, how will we run it?	• What should the output look like? • Draft the chart we expect to produce (with dummy data)	• Who is responsible? • What steps will that person need to take? • When can we expect the results?

Plan analyses - Where is the drop-off happening?

42% of visitors drop off between entering their tracking ID and viewing shipment details



Plan analyses - Diagnosing Tracking-Page Drop-off

Problem statement: *Diagnose why 42% of visitors drop off between entering their tracking ID and viewing shipment details on the tracking page, and identify the highest-impact fix to cut this drop-off to under 15% within 3 weeks*

Issue	Hypothesis	Analysis	Sources of insight	End product	Timing /steps / responsibility
Customer Behavior: Customers may be mistyping or forgetting their tracking ID at this step.	High ID-submission error rate is causing early exits (customer-behavior cause).	• Measure the ID-submission error rate. • Segment errors by order channel (email link vs. manual entry).	• Error logs for manual entry vs automated fill • Support tickets tagged "tracking issue"	**Error breakdown**: share of drop-off explained by ID errors.	**Data team:** Pull error logs (2 days); analyze by channel (1 day)
Frontend/UX: The results page may be confusing or slow to load on mobile.	Mobile users are exiting due to a cluttered or slow results page (frontend/UX cause).	**Review session recordings for this step.** • Compare drop-off rate by device type	• Session-recording tool • Funnel analytics split by device	**Device comparison** • Drop-off rate by device type. • Annotated screenshots of friction points.	**FDE + design:** Review session recordings (3 days) Compare desktop vs. mobile funnel (1 day)
Backend/Technical: The shipment-details page may be slow to load after ID submission.	Slow backend response time is causing users to abandon before the page loads (backend/technical cause).	**Benchmark performance at this step:** • Page-load time after ID submission • API response time • Server error rate during peak hours	• Server response-time logs • Page-load monitoring dashboard • Peak-hour traffic reports	**Load-time benchmark** • Response time split by traffic	**Backend eng + FDE:** Pull response-time logs (2 days); benchmark load time (1 day)

THE 15-CLASS FDE JOURNEY

Four phases. Fifteen classes.

PHASE 1 DISCOVER

- C1 Stakeholder Discovery
- C2 Workflow + Problem Structuring
- C3 KPI + Scope

YOU ARE HERE — C3

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- C5 SQL + APIs + Files
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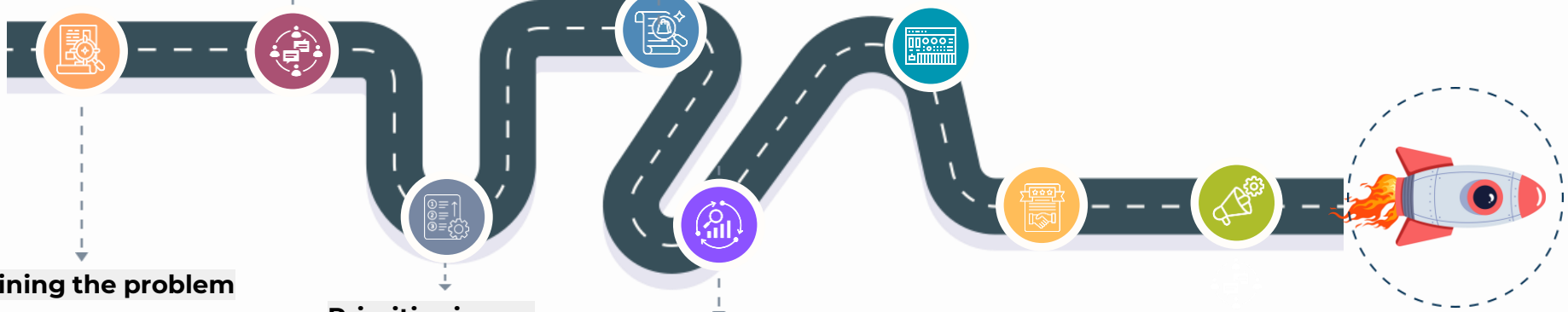
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Conduct analyses

What's the quickest and simplest way to ***test the hypothesis?***



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Conduct Analyses

Unlocking Insights: The Art and Science of Analysis

Quantitative analysis

'Quantitative analysis' turns numbers into insights, revealing the '***what***' with precision



Data segmentation and analysis



Technical analysis *(e.g. API response time)*



Financial Metrics *(e.g. cart conversion, average order value)*



User Metrics *(e.g. Monthly active users, repeat users)*



Performance Metrics *(e.g. customer success, tickets)*

Qualitative analysis

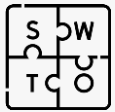
'Qualitative analysis' uncovers the '***why***' behind the data, giving meaning to the numbers



Stakeholder Interviews



Benchmarking/ Case Studies



SWOT/PESTLE Analysis



Customer Journey & User Experience Mapping



Focus Groups & Surveys

Mind Traps: How Biases Sabotage Smart Problem-Solving (1/3)

Scenario: *Your study group is tackling a case study where a website needs to reduce page drop-off to 15% in 3 weeks. As you work through the case, cognitive biases start influencing the team's decision-making*

Confirmation bias



"**Sticking** to own beliefs"

Anchoring bias



"**First info** sticks and drives all **decision making**"

Availability bias



"Going with the **easiest or most familiar** option"

Over-optimism bias



"**Underestimating challenges** due to excessive confidence "

Mind Traps: How Biases Sabotage Smart Problem-Solving (2/3)

Scenario: *Your study group is tackling a case study where a website needs to reduce page drop-off to 15% in 3 weeks. As you work through the case, cognitive biases start influencing the team's decision-making*



Mind Traps: How Biases Sabotage Smart Problem-Solving (3/3)

Scenario: *Your study group is tackling a case study where a website needs to reduce page drop-off to 15% in 3 weeks. As you work through the case, cognitive biases start influencing the team's decision-making*

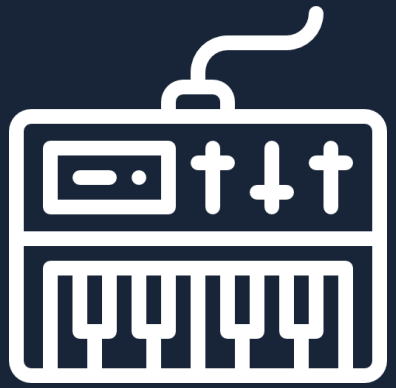
Escalation of commitment / Commitment Bias



COMMITMENT

"**Keep investing in a decision or action,** even when it's facing negative outcomes "

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Synthesize Findings

Pause and reflect

Structuring the problem

How can we break the problem into MECE elements for maximum insight?

Plan analyses

What data do we need, ***how*** will we collect it, ***who*** is ***responsible, and what's*** *the* ***timeline?***

Synthesize findings

Define the "so whats?"



Defining the problem

What's the real issue?
Are we solving the root cause, or just a symptom?

Prioritize issues

What key questions, if answered, will drive the most impact?
Don't get lost in the noise.

Conduct analyses

What's the quickest and simplest way to ***test the hypothesis?***

So What? The path from Situation to Resolution

	Description	Example
Situation (Reason for action)	Briefly describe the current state	"ABC Retail Co. operates a chain of brick-and-mortar stores focused on providing high-quality in-store customer experiences."
Complication (Why)	What opportunity has arisen? What threat has emerged?	"Increased competition from online retailers has driven down foot traffic, while the rise of experiential retail presents an opportunity to enhance in-store customer engagement and differentiate from online competitors"
Resolution (Answer)	What should we do?	"ABC Retail Co. should develop and implement a customer-centric training program for store staff, paired with new in-store experiences, to attract and retain more customers"

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Craft Recommendations

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Crafting Recommendations:
Turning Insights into Actionable Plans

Actionable recommendations	• Provide clear actions for achieving the desired outcome
Establish a success metric / KPIs	• Clean cut metrics (key performance indicators) and tracking mechanism
Pragmatic Action Plan	• Identify relevant initiatives • Detail the sequence, timing, and activities • Assign clear ownership for each initiative • Establish Check mechanism for repeatability • SOPs creation
Key Success Factors	• Identify challenges and factors critical for success

Defining success metrics

Leading indicators

When to use?

- During the testing phase or piloting during a test window or test users
- Early warning to see failure

Examples

- Weekly tracking ID funnel rate in the A/B test
- % of test users revisiting within minutes
- Checkout page load time

Lagging indicators

When to use?

- Top-most project metric defined in SMART problem statement
- Why the project was given

Examples

- Monthly tracking ID drop-off rate
- Repeat customers over 6 months
- Checkout-related support complaints

Leading indicators tell you within days whether the fix is working; lagging indicators confirm the business result weeks later.

Scenario - Warehouse Pick-and-Pack

Scenario: Your client is an e-commerce retailer that has noticed a drop warehouse efficiency and orders are taking too long to fulfill. Identify the reasons and recommend solutions

How to define the problem statement?

Specific

Figure out why the warehouse pick-and-pack process is seeing **an average of 12 minutes per order** ~~and orders are taking too long to fulfill~~ so ~~we should try to speed things up~~ and **cut pick-and-pack time** to **6 minutes per order** without changing staff **within the next 4 weeks**

Measurable

Actionable

Realistic

Time-bound

Plan analyses - Diagnosing Slow Warehouse Pick-and-Pack

Problem statement: *Diagnose why average pick-and-pack time is 12 minutes per order and identify the highest-impact fix to reduce the time to 6 minutes per order within 4 weeks*

Issue	Hypothesis	Analysis	Sources of insight	End product	Timing /steps / responsibility
Process: Pickers may be taking inefficient routes through the warehouse.	**Manual routing is causing extra walking time between items (process cause).**	• Map current pick routes for a sample of orders. • Measure average distance walked per order.	• Warehouse layout map • Time-motion observations	**Route-efficiency analysis** showing potential time savings from optimized paths.	**Ops team:** Shadow pickers (2 days); map routes (1 day)
Technology: Pickers may be validating wrong order ID and packing wrong items	Manual SKU verification is adding time per item (technology cause).	**Time manual verification steps per item.** • Compare to scanning benchmarks from similar warehouses.	• Vendor benchmarks for barcode-scanning setups	**Time-savings estimate** • from introducing barcode scanning. • Basis for the recommendation in the next section.	**FDE:** Benchmark research (1 day)
People: Newer pickers may be slower due to limited training.	Inconsistent training is causing a wide spread in pick times across staff (people cause).	**Compare pick times by:** • Staff tenure / training level • Identify gap between new and experienced staff • Check if the gap narrows after standardized onboarding	• Staff tenure records • Individual pick-time logs • Onboarding / training records	**Training-gap analysis** • showing whether there is a significant difference in pick-time based on staff tenure	**Data team:** Pull tenure data (1 day); analyze pick-time spread (1 day)

Crafting Recommendations:
Speeding Up Warehouse Pick-and-Pack



Actionable recommendations

- Initiative: Introduce barcode scanning at each pick station to speed up warehouse pick-and-pack



Establish a success metric

- **Success Metric:** Cut average pick time from 12 minutes to 6 minutes per order
- **Plan Commitment:** Ensure the warehouse ops manager signs off on the pilot



Pragmatic Action Plan

- **Initiatives:** Pilot barcode scanners at pick stations
- **Sequence & Timing:** Run a 2-week pilot in one warehouse, then scale to all sites within 4 weeks
- **Ownership:** Warehouse ops lead runs the pilot, shift supervisors handle daily use
- **Repeatability:** Keep a manual-entry fallback if scanners are down during testing
- **SOPs Creation:** Document the new pick-station workflow



Key Success Factors

- **Challenges:** Barcode label print quality issues and staff resistance to a new scanning step
- **Critical Factors:** Clear training, a manual fallback for scan failures, and tracking pick-time and error-rate closely during the pilot

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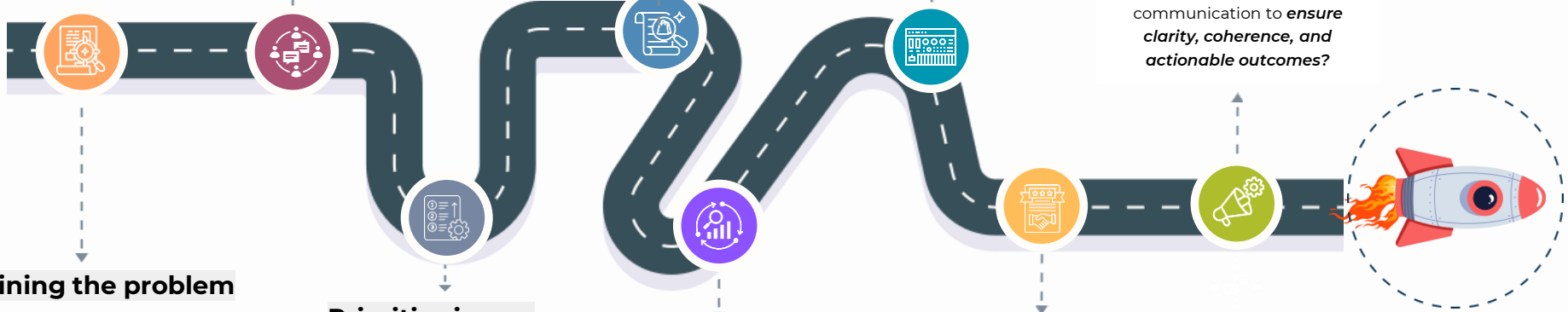
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Communicate

How can we structure communication to ***ensure clarity, coherence, and actionable outcomes?***



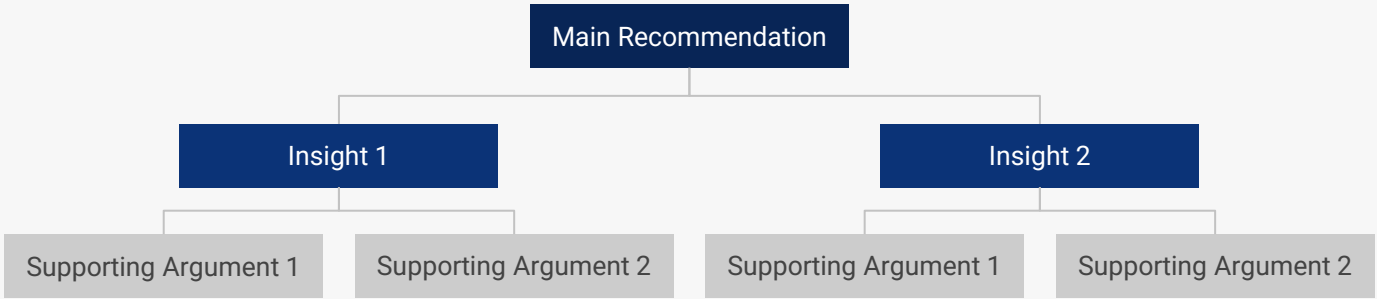
The Power of Effective Communication: Maximizing Impact with the Pyramid Principle

Why is effective communication important?

1 problem statement	3 problem statement	7 problem statement	12 problem statement
			
You (FDE)	Director of strategy	CTO	Founder/CEO

The "**return on investment**" for each word you tell a CEO/CTO has to be very high, as they have only limited time and attention - "**Density per word should be high**"

Pyramid principle for effective communication



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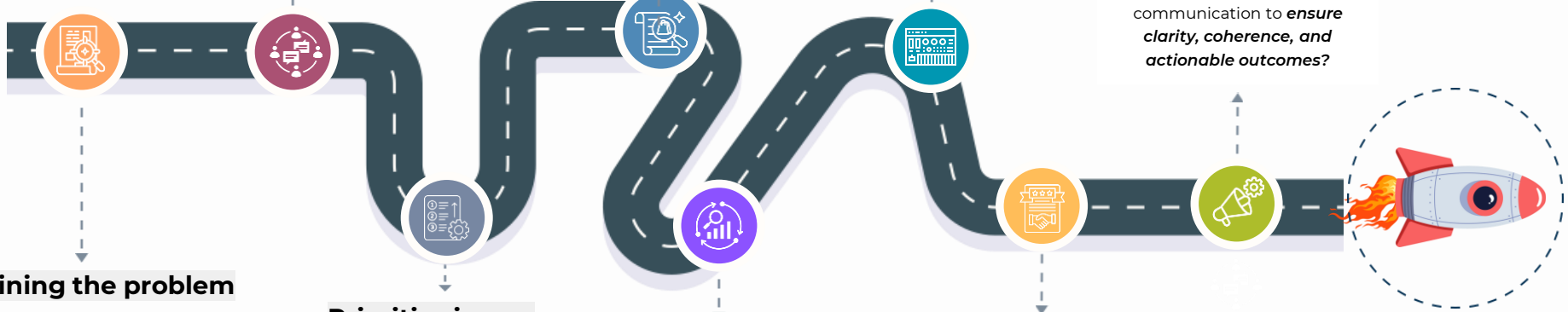
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Scoping

Problem statement worksheet - Template

Basic question to be resolved

- *Defines what must be resolved to **deliver high impact***
- *The question should be **SMART**: specific, measurable, action-oriented, relevant, and time-bound*

Context

Sets out the situation and complication facing the client - e.g., industry trends, relative position in the industry

Criteria for success

Defines success for the project. Must be shared by client and team, and must Include relevant qualitative and quantitative measures - e.g., impact and impact timing, visibility of improvement, client mindset shifts

Scope of solution space

Indicates what will and will not be Included In the study - e.g., international markets, research and development activities, uncontrolled corporate costs

Timeline for resolution

How quickly is the answer needed?

Constraints within solution space

Defines the limits of the set of solutions that can be considered - e.g., must involve organic rather than inorganic growth

Stakeholders

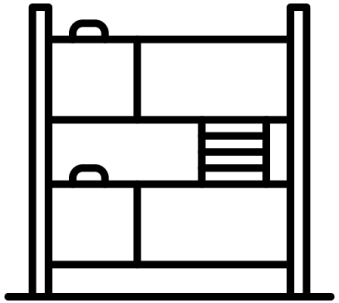
Identifies who makes the decisions and who else could support (or derail) the study - e.g., CEO, division manager, SBU manager, key outside influencers

Key sources of insight

Identifies where best-practice expertise, knowledge, and engagement approaches exist (Internal and client) - e.g., practice experts, EM guides, practice databases

Problem statement worksheet - Retail inventory sync (1/2)

Finalized Problem Statement



Problem statement: Fix the warehouse-to-website inventory sync so the retailer stops overselling items during flash sales. The objective is to reduce the sync delay to under 30 seconds and eliminate oversold SKUs, without any site downtime, before the next scheduled flash sale in three weeks

Questions?

THE 15-CLASS FDE JOURNEY

Four phases. Fifteen classes.

PHASE 1 DISCOVER

- C1 Stakeholder Discovery
- C2 Workflow + Problem Structuring
- C3 KPI + Scope

YOU ARE HERE — C4

PHASE 2 DATA

- C4 Client Data
- C5 SQL + APIs + Files
- C6 Data Quality
- C7 Business Data Model
- C8 Dependable Pipeline

PHASE 3 BUILD + AI

- C9 Product Wireframe
- C10 Agentic Coding
- C11 End-to-End Build
- C12 AI Agents
- C13 Multi-Agent + Evaluation

PHASE 4 SHIP

- C14 Deploy + Observe
- C15 Client Review + Defence